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Editor-in-Chief
Discover Networks
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Dear Editor,

I am pleased to submit our manuscript, “Beyond Node Degree: Identity Collapse and Edge-Weight Concentration in Graph-Based Network Intrusion Detection,” for consideration as a Research Article in Discover Networks.

Graph-based network intrusion detection almost universally represents attacker behavior through node-centric structural features – degree, centrality, embeddings from graph neural networks – which implicitly assume that an attacker’s identity is visible as a distinct node in the host-communication graph. This manuscript identifies a common, realistic condition under which that assumption fails: network address translation (NAT) and similar perimeter-aggregation mechanisms, which collapse many real hosts onto a small number of externally visible addresses. Using real, labeled flow data from the CICIDS2017 benchmark, we show that once an attack is observed through such a collapsed vantage point, standard node-degree features become statistically indistinguishable from benign traffic (Mann–Whitney $p = 0.12\text{--}0.83$), while a simple edge-weight concentration feature separates attack from benign traffic perfectly and robustly across window sizes, thresholds, classifier families, and random seeds. We further show that a compact, ten-dimensional feature set built on this principle matches the detection accuracy of the standard 78-dimensional flow-feature baseline at a fraction of the computational cost – a result with direct relevance to resource-constrained network operators such as small and medium enterprises and IoT/edge deployments.

We believe this work is well suited to Discover Networks given the journal’s focus on technological and smart networks, network security, and infrastructure resilience, and its stated interest in work connecting to the Industry, Innovation and Infrastructure Sustainable Development Goal. The manuscript includes a closed-form theoretical explanation for the phenomenon, an ablation study identifying which structural features carry the signal, robustness checks across multiple analysis choices, and an explicit, transparent discussion of the study’s current limitations – most notably that empirical validation is presented for a single attack archetype, with the specific follow-up validation required to generalize the claim stated plainly in the manuscript.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. I have no conflicts of interest to declare, and this research received no external funding. All data used are drawn from the publicly available CICIDS2017 benchmark, and all analysis code, results, and figures underlying the manuscript are made available at the GitHub repository cited in the manuscript’s Data and Code Availability statement, to support full reproducibility.

Thank you for your time and consideration. I look forward to the editorial process and welcome

any questions.

Sincerely,

Md Hasibuzzaman